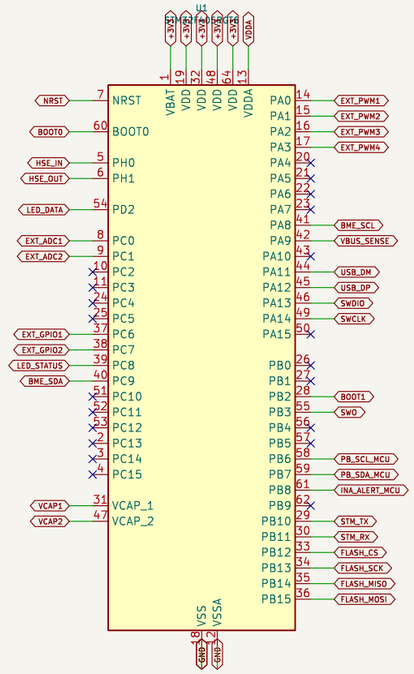
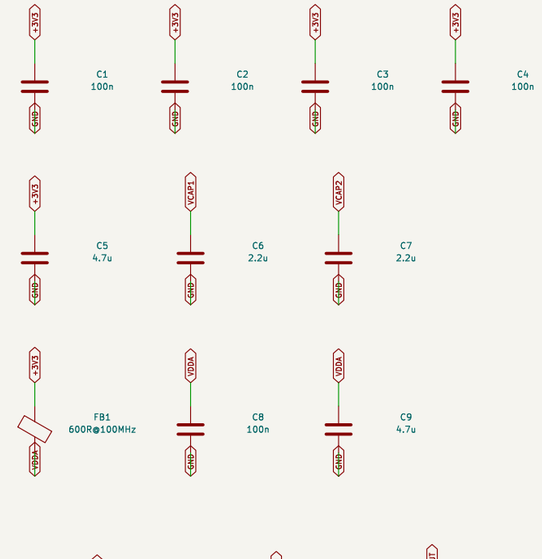
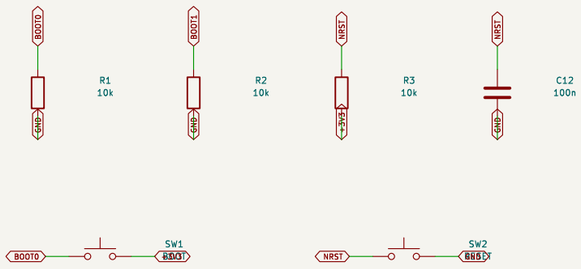
Core: HSE8MHz; VDDA filtered; VCAP2x2.2uF. No motor / Lumen / Navigator sensors.







Power_USB

File: Power_USB.kicad_sch

Peripherals

File: Peripherals.kicad_sch

Interfaces

File: Interfaces.kicad_sch

HYDRA / Proteus + HydroVinci

Sheet: /

File: HYDRA_MAIN_V1.kicad_sch

Title: HYDRA MAIN BOARD V1 — HYDRA_MAIN_V1

Size: A3	Date: 2026-10-27	**Rev: A1 COMPACT**
KiCad E.D.A. 9.0.9		Id: 1/4

Input 5V only 4.75–5.25V. Schottky OR drops voltage; +5V_SYS is not a precision 5V rail.

PB_5V
GND
J1
5V POWER IN
PB_5V
F1
0.50A hold
+5V_SYS
D1
PB_FUSED
VBUS_USB
F2
0.50A hold
+5V_SYS
D2
USB_FUSED
+5V_SYS
GND
C13
4.7u
+5V_SYS
IN
OUT
GND
+3V3
AP7361C-33E-13
+3V3
GND
C14
4.7u
+5V_SYS
GND
C15
100n
USB_CC1
R4
5.1k
USB_CC2
R5
5.1k
GND
VBUS_USB
C16
100n
GND
VBUS
CC1
CC2
D–
D–
D+
D+
SHIELD
GND
SBU1
SBU2
USB
A4
A5
B5
A7
B7
A6
B6
A8
B8
S1
A1
VBUS_USB
USB_CC1
USB_CC2
USB_DM_EXT
USB_DM_EXT
USB_DP_EXT
USB_DP_EXT
GND
GND
VBUS_USB
USB_DP_EXT
USB_DM_EXT
U5
USB_DP_EXT
USB_DM_EXT
GND
USB_DM_EXT
R6
22
USB_DP_EXT
R7
22
VBUS_USB
R8
33k
VBUS_SENSE
VBUS_SENSE
R9
82k
GND
GND
#FLG1
PWR_FLAG
+5V_SYS
#FLG2
PWR_FLAG
VBUS_USB
#FLG3
PWR_FLAG

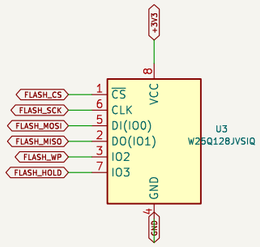
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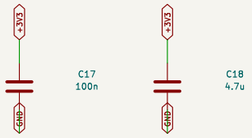
Sheet: /Power_USB/
File: Power_USB.kicad_sch

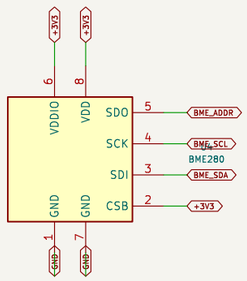
Title: HYDRA MAIN BOARD V1 — Power_USB

Size: A3	Date: 2026-10-27	**Rev: A1 COMPACT**
KiCad E.D.A. 9.0.9		Id: 2/4

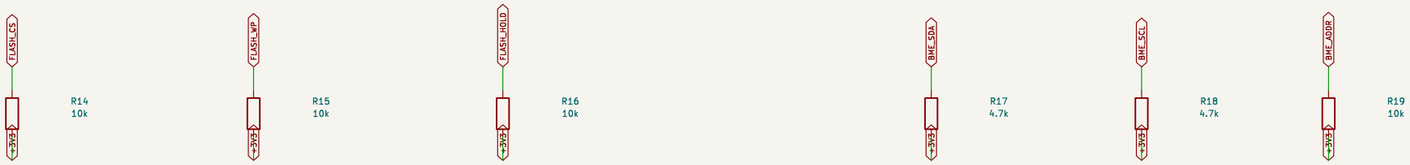
BME280 I2C3 0x77. Flash SPI2. STATUS and DATA are firmware-driven GPIO LEDs.

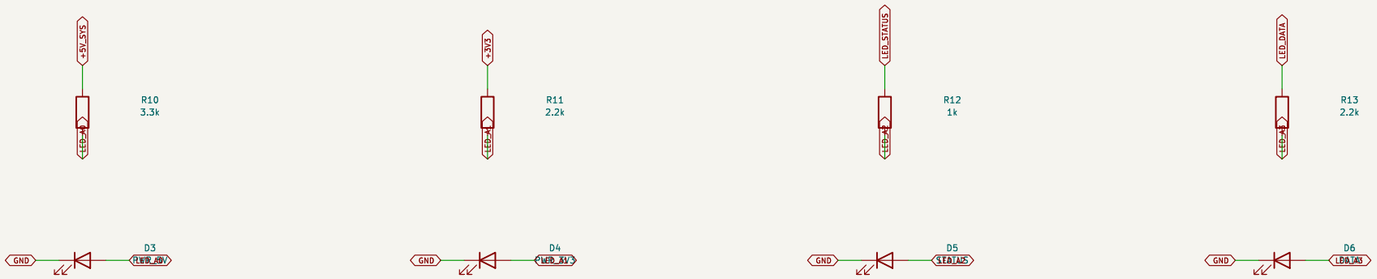












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Sheet: /Peripherals/
File: Peripherals.kicad_sch

Title: HYDRA MAIN BOARD V1 — Peripherals

Size: A3	Date: 2026-10-27	**Rev: A1 COMPACT**
KiCad E.D.A. 9.0.9		Id: 3/4

All cable pin numbering: connector mating face, confirm latch orientation. INA pullups only here; no hotplug.



J2
POWER I2C ALERT



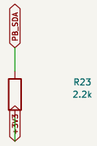
R20
47



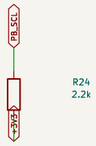
R21
47



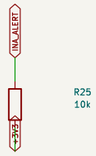
R22
1k



R23
2.2k



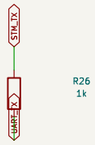
R24
2.2k



R25
10k



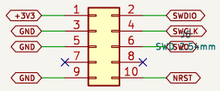
J3
RPi UART 3V3



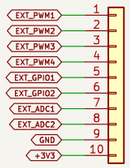
R26
1k



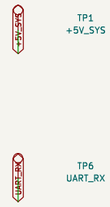
R27
1k



SW1
SWD_5x2_1.27mm



J4
8 GPIO V2 LOGIC



TP1
+5V_SYS

TP2
+3V3

TP3
GND

TP4
NRST

TP5
UART_TX

TP6
UART_RX

TP7
PB_SDA

TP8
PB_SCL

TP9
INA_ALERT

TP10
BOOT0

H1
M2.5 NPTH

H2
M2.5 NPTH

H3
M2.5 NPTH

H4
M2.5 NPTH

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Sheet: /Interfaces/
File: Interfaces.kicad_sch

Title: HYDRA MAIN BOARD V1 — Interfaces

Size: A3	Date: 2026-10-27	Rev: A1 COMPACT
KiCad E.D.A. 9.0.9		Id: 4/4